Security Analysis of Autonomous Cyber -Physical Systems Using Deep-Learning

ABSTRACT

Secure and autonomous cyber-physical systems (CPS) play a critical role in safety-critical domains such as intelligent transportation, industrial automation, and smart infrastructure. However, increasing connectivity and autonomy expose these systems to cyber-physical attacks that can compromise safety and reliability, making accurate detection and mitigation essential for resilient autonomous operations. This study presents a **deep learning–based security-aware learning and control framework** to enhance the resilience of autonomous CPS by integrating systematic system modeling, secure state estimation, and **adversarial-aware deep learning techniques** to address sensor attacks, false data injection, and control manipulation. The proposed framework employs a **hybrid CNN–LSTM architecture**, where CNN layers extract spatial and protocol-level features and LSTM layers capture temporal attack patterns in CPS data. Robust control techniques are combined with deep learning models to improve system stability under uncertainty and adversarial conditions. The framework is evaluated using the **BACnet Intrusion Detection Dataset**, which contains building automation system traffic with normal and malicious cyber-physical behaviors. Performance is assessed using standard metrics such as attack detection accuracy, control stability, and system robustness. Experimental results demonstrate that the proposed framework achieves improved detection performance and enhanced resilience compared to conventional GAN-based and control-only approaches, contributing to safer and more trustworthy cyber-physical infrastructures.

Keywords: Cyber-Physical Systems (CPS), Deep Learning, CNN–LSTM Architecture, Intrusion Detection, Zero-Day Attack Detection, Secure State Estimation, Robust Control, BACnet Intrusion Detection Dataset, Cyber-Physical Security

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